

CXR GraphQL Lab Manual

SW.15 — Apollo Gateway + federated mock services

CXR Ops Lab — `cxr-ops-lab`

2026-05-31

Abstract

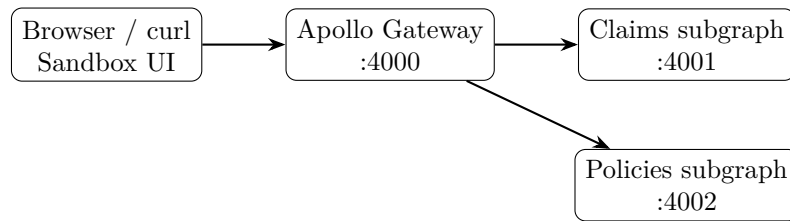
SW.15: Apollo **Gateway (:4000)** composes two subgraphs — **claims (:4001)** with CXR-shaped **Claim** types and **policies (:4002)** with mock **Policy** lookups. Standalone lab; **Claim Studio (:3000)** remains REST.

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Chapter 1

Architecture



Chapter 2

CXR types (syllabus)

2.1 Claim (claims subgraph)

```
type Claim {  
  id: ID!  
  status: String!  
  summary: String  
}
```

Fixtures: demo-1 (status ok), demo-2 (status review).

2.2 Policy (policies subgraph)

```
type Policy {  
  id: ID!  
  claimId: ID!  
  code: String!  
  description: String  
}
```

policyForClaim(claimId: "demo-1") → CXR-POL-OK.

Chapter 3

Procedure

3.1 Start stack

```
cd cxr-ops-lab
./scripts/18-graphql-up.sh
./scripts/18-graphql-smoke.sh
```

3.2 Browser golden path

1. Open <http://localhost:4000/graphql> (Apollo Sandbox).
2. Paste query from `lab/graphql/query-golden.graphql`.
3. Expect `claim.status = ok` and `policyForClaim.code = CXR-POL-OK`.
4. Screenshot `evidence/SW15-graphql-sandbox-*.png` (optional).

Chapter 4

File inventory

File	Purpose
<code>compose.graphql.yaml</code>	Docker Compose for gateway + subgraphs
<code>lab/graphql/Dockerfile</code>	Node 20 image
<code>lab/graphql/claims-subgraph.mjs</code>	Federated claims service
<code>lab/graphql/policies-subgraph.mjs</code>	Federated policies service
<code>lab/graphql/gateway.mjs</code>	IntrospectAndCompose gateway
<code>lab/graphql/schema.graphql</code>	Composed schema reference
<code>lab/graphql/query-golden.graphql</code>	Golden path query
<code>scripts/18-graphql-up.sh</code>	Build + start
<code>scripts/18-graphql-smoke.sh</code>	Subgraph + federated probe
<code>evidence/SW15-graphql-verify-2025-05-31.md</code>	© 2025-05-31

Chapter 5

Claim Studio vs GraphQL

App	URL	Protocol
Claim Studio	:3000	REST / Next.js API
GraphQL gateway	:4000	GraphQL (lab only)

Chapter 6

After SW.15

SW.16 gRPC sketch.

Chapter 7

Live adjustments (federation boot)

- Subgraph SDL passed through `parse()` from `graphql` — required for `buildSubgraphSchema` (string alone caused `definitions is not iterable`).
- Gateway `IntrospectAndCompose` waits for healthy subgraphs on Docker network URLs.
- Healthchecks on `:4001/:4002` before gateway `:4000` starts.
- Not wired to Claim Studio `:3000` — REST remains production spine; GraphQL is syllabus lab only.

Chapter 8

UI evidence — Apollo Sandbox (:4000)

Query Sw15GoldenPath: federated claim(demo-1) + policyForClaim.

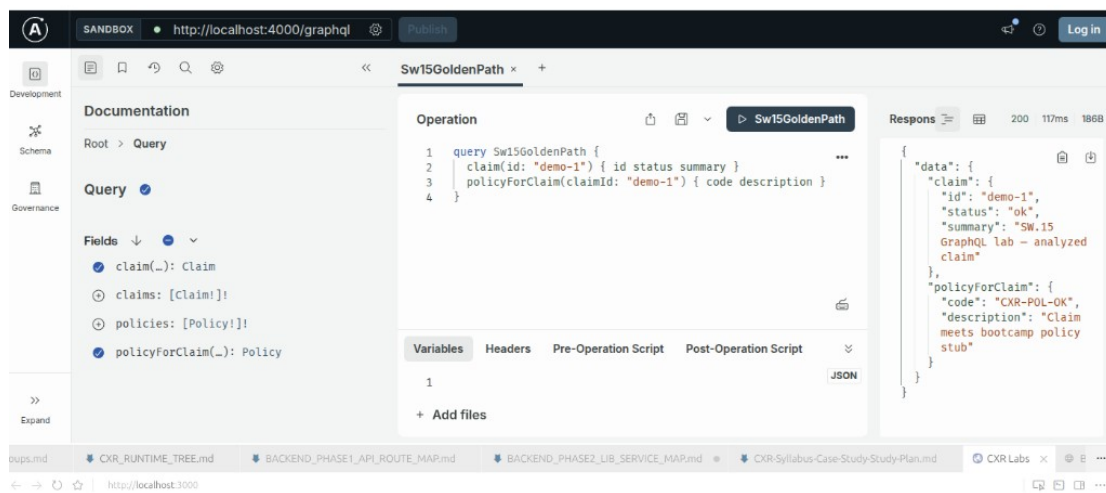


Figure 8.1: Apollo Sandbox — `status: ok, CXR-POL-OK` (SW.15 evidence).